H Roll No. .....

TCS–201

B. TECH. (SECOND SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

PROGRAMMING FOR PROBLEM SOLVING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Explain the concept of a row major and column major form of a 2-D array with an example showing the evaluation performed by the compiler to access an element of any array.

(CO1, CO2)

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(ii) Develop a C program to read the integer elements of a matrix of size M-row and N-columns from the user and then swap the elements of first column with the elements of the last column. Display the modified matrix to the output screen. (CO1, CO2)

OR

(b) Draw a flowchart and write a C program to read two square matrices of order M × M and then find the remainder by dividing the elements of the first matrix with the elements of the second matrix and store the result in the third matrix. Print the resultant matrix to the screen. (CO1, CO2)

2. (a) (i) Illustrate the concept of 1-D character array to store a string in C. What is the significance of a Null character ? Give a snippet of a C code to read a string containing spaces from the keyboard. (CO1)

(ii) Design a user defined function (UDF) to return the modified string by replacing all the white spaces contained in it with a '#' character. Implement a C program to read a sentence from the user and display the modified string. (CO1)

OR

(b) Demonstrate and briefly explain with a snippet of C code the use of library functions to find the length of a string, copy a string to another string, and compare two strings. (CO1)

3. (a) (i) Illustrate concept of a pointer in C. Write a program in C to demonstrate the meaning of call by reference. (CO3)

(ii) Develop a C program to read a string and print the number of vowels and consonants present in it using a pointer to the console. (CO3)

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OR

(b) (i) Design a UDF to reverse a string passed to it using a pointer and store it back into the same. Write a C program to read a string in the main program and display it in reverse to the console. (CO3)

(ii) Develop a C program to find the maximum and minimum element of an array of size N using a pointer. (CO3)

4. (a) Programmer can optimize the usage of runtime memory by implementing the dynamic memory allocation function. Illustrate with an example of the use of functions to allocate, resize and free the memory. (CO3)

OR

(b) Illustrate briefly with a snippet of code the meaning of the following : (CO3)

(i) Null Pointer

(ii) Dangling pointer

(iii) Void pointer

(iv) Wild pointer

5. (a) Predict the output of following program code : (CO1, 2, 3)

Note : All the given C program are free of syntax errors. Assume appropriate tokens if any missing.

(i)

```
#include<stdio.h>
void main ()
{
  char str [] = "peace";
  char *s = str;
  printf ("%s\n", ++s + 1);
}
```

(ii)

```
#include<stdio.h>
void main()
{
char s[] = "Program/0ming is fun";
printf("\n%s", str + 3);
}
```

(iii)

```
#include<stdio.h>
```


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```
void main()
{
int *ptr []= {11, 22, 33, 44};
int i = 1;
while (i < 4)
{ printf("%d\t", * (ptr + (i -1)));
        i++;
}
}
```

(iv)

```
char fun 1 (char s1 [])
{
return (s1 [0]);
}
char fun2 (char s2 [])
{
return (s2 [4]);
}
void main()
{
printf("%c%c", ful 1 ("Graphic"),
fun 2 ("Grafest"));
}
```


(v)

```
#include<stdio.h>
include<string.h>
void main ()
{
  char *str={''Secretes are hidden
  like streasures");
  printf("%s+%d",
                str+12,strlen(str+12));
}
```

Or

(b) Design a UDF using pointer to reverse elements of 1-D array passed to it. Write a C program to read the N numbers of the 1-D array in the main program and then print the modified array after calling the function. (CO1, 2, 3)

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